

Ahmad Fahim Basheer Ahmad

ahmadfahim2k@gmail.com | +44 7818436241 | Liverpool, United Kingdom | [linkedin.com/in/ahmadfahim2k](https://www.linkedin.com/in/ahmadfahim2k)
github.com/ahmadfahim2k | portfolio-self-pi-31.vercel.app

Data Scientist with hands-on experience in NLP, deep learning, RAG pipelines, and end-to-end ML systems, currently completing a Master's in Advanced Data Science and AI at the University of Liverpool. Backed by 3+ years as a full-stack developer in the healthcare and pharmaceutical domain, bringing strong software engineering practices to ML pipeline development and deployment.

Education

Master of Advanced Data Science and Artificial Intelligence

[University of Liverpool](#)

Sep 2025 – Sep 2026 | Liverpool, United Kingdom

- Engineered disease-spreading simulation in C and implemented parallelization and optimization to achieve 28x speedup with OpenMP and 31x speedup with MPI
- Collaborated on a group project to build a full-stack financial crisis analysis tool, analysing 3,158 US equity tickers across crisis periods using Next.js, FastAPI, and LLM-generated insights
- Designed and analyzed algorithms for sorting and optimization problems, including pancake-sorting strategies and Hamming-code based encoding schemes

Bachelor of Technology in Computer Science and Engineering

[B.S. Abdur Rahman Crescent Institute of Science and Technology](#)

Jul 2018 – May 2022 | Chennai, India

- First Class with Distinction
- CGPA 8.73/10

Experience

[Virtusa Consulting Services](#)

Associate Engineer

Jun 2022 – Aug 2025 | Chennai, India

- Migrated three healthcare front-ends to a custom library, completing the project in 12 months
- Built reusable React components with Tailwind CSS and Recoil in Storybook, achieving 100% UI parity post-migration
- Fixed over 30 bugs post-migration, collaborating with QA and earning Star Performer of the Month
- Worked within Agile/Scrum teams, using JIRA to manage sprints, track deliverables, and coordinate cross-functional work
- Advanced from junior developer in a two-person team to sole developer on a dedicated project, owning requirements gathering, feasibility analysis, feature demos, and sprint delivery
- Conducted technical interviews for prospective hires and mentored new joiners in React to ensure readiness for client-facing project assignments

Projects

[Spectral Clustering Analysis : Benchmarking Graph Clustering on Real-World Networks](#)

An interactive benchmark comparing five clustering algorithms across transport, biology, and social network datasets to expose when spectral methods fail and why standard metrics can mislead

June 2026 – September 2026

- Implemented five clustering algorithms from scratch in Python (NumPy, SciPy) for direct eigenvector-level diagnostics
- Diagnosed a 726-of-727-node collapse in unnormalised spectral clustering on a 726k-edge airport network, tracing it to a single low-degree node's eigenvector position
- Exposed a cluster-validity metric scoring an identical clustering from 0.019 (“excellent”) to 6.27 (“terrible”), and introduced Modularity as a more reliable alternative
- Built and deployed an interactive results explorer with Next.js, React, and D3.js, hosted on Vercel with CI/CD via GitHub auto-deploy

github.com/ahmadfahim2k/spectral-clustering-analysis

[Stock Analyzer : Financial Crisis Analysis Tool](#)

Full-stack web application for analysing US equity performance across major historical financial crises

Feb 2026 – Jun 2026

- Developed a FastAPI backend serving OHLC data for 3,158 US equity tickers (S&P 500, NASDAQ, NYSE, Forbes 2000) spanning 1970–2022
- Integrated Groq LLM to generate natural language summaries of computed metrics for a stock and time period

- Built an interactive Next.js frontend with Recharts visualisations for comparing stock performance across the Dot-Com crash, 2008 Financial Crisis, and COVID-19
- Deployed the full stack to Azure Container Apps with a GitHub Actions CI pipeline validating Docker builds on every push, catching a build failure before deployment

github.com/ahmadfahim2k/stock-analyzer-group-project

ShoeBot : E-Commerce Chatbot

LLM-based shoe shopping assistant that handles product search, FAQs through a conversational interface

Apr 2026

- Implemented semantic router using HuggingFace to classify user intent and dispatch queries to focused models
- Created a natural language to SQL pipeline using Groq to query a web-scraped Flipkart product catalog
- Integrated FAQ retrieval using RAG with ChromaDB and deployed the chatbot as a Streamlit application

github.com/ahmadfahim2k/shoebot-genai

Vehicle Damage Prediction

Deep learning system that classifies vehicle damage from images into 6 categories

Jan 2026

- Trained a vehicle damage image classifier using transfer learning with ResNet50 in PyTorch, achieving ~81% validation accuracy across 6 damage categories
- Tuned hyperparameters with Optuna and applied data augmentation to improve model generalization
- Served the model via a Streamlit app and FastAPI REST API for real-time inference on uploaded vehicle images

github.com/ahmadfahim2k/dl-project-vehicle-damage-detection

Healthcare Premium

Regression model that predicts annual health insurance premiums from user inputs

Aug 2025

- Developed an end-to-end ML pipeline with XGBoost to predict healthcare insurance premiums, achieving R^2 of 0.998 on unseen data
- Applied age-stratified modelling with separate models for young and older cohorts after error analysis revealed distinct pricing patterns
- Constructed a composite medical risk score from 4 chronic condition indicators to improve accuracy
- Launched a Streamlit app that routes user inputs to age-stratified models and returns premium estimates

github.com/ahmadfahim2k/ml-project-healthcare-premium

Skills

Languages — Python, JavaScript, C, SQL

Frontend — React, Tailwind CSS, Recoil, Storybook, Next.js, D3.js

Backend — Node.js, Django, FastAPI

Data Analysis — Pandas, NumPy, SciPy, Matplotlib

ML / AI — PyTorch, Scikit-Learn, XGBoost, LangChain, ChromaDB, HuggingFace, Optuna, RAG, NetworkX

Tools — Git, Streamlit, Docker, Azure, Vercel

Practices — Agile, Scrum, JIRA, CI/CD

Certificates

[Oracle: Certified Java Programmer](#)

[CodeBasics: Machine Learning](#)

[CodeBasics: Deep Learning](#)

[CodeBasics: Math and Stats for AI / ML](#)